## ④ Model considering the overall market and multiple penalties

**[1. Construct two types of demand has been moved to pg3.]**

### 1. Core model: City selection + vehicle deployment + multiple penalties

$$\min_{y,q,O,U} \quad \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle costs}} + \frac{1}{|S|}\sum_s \sum_c \left[ \underbrace{c_c^+ O_c^s}_{\text{Remaining capacity}} + \underbrace{c_c^- U_c^s}_{\text{Foregone contribution margin}} \right].$$

$$\min_{y,q,R,O,U,H,L} \quad \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle costs}} + \frac{1}{|S|}\sum_s \sum_c \left[ \underbrace{c_c^+ O_c^s}_{\text{Remaining capacity}} + \underbrace{c_{c,H}^- H_c^s}_{\text{Loss from switching to human-driven taxis}} + \underbrace{c_{c,L}^- L_c^s}_{\text{Other unserved loss}} \right].$$

In $c^+, c^-$, $+, -$ distinguish excess and shortage, rather than the signs of costs. All demand, throughput, and costs use **the same evaluation period.** ※ I suppose that setting the evaluation period is needed, and the model determines the city/vehicle deployment to apply during that period.

### 2. Constraints *[Moved from page 3 to page 1 in this revision.]*

$$\begin{aligned} 0 \le q_c \le \bar{q}_c y_c \qquad & \forall c, \\ \sum_c q_c \le Q, & \\ \kappa_c q_c - M_c p_c^s = O_c^s - U_c^s \qquad & \forall c, s, \\ y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0}, & \\ O_c^s, U_c^s \ge 0. & \end{aligned}$$

| Constraint | Meaning |
|---|---|
| $0 \le q_c \le \bar{q}_c y_c$ | Zero vehicles if the city is not entered; up to the city-specific limit if it is entered |
| $\sum_c q_c \le Q$ | Total vehicle limit |
| ~~$y_c \le a_c$~~ | ~~Enter only cities where robotaxi operation is feasible at the planning stage~~ |
| ~~$R_c^s + O_c^s = \kappa_c^s q_c$~~ | ~~Total service capacity = used capacity + remaining capacity (unit: trips)~~ |
| ~~$R_c^s + U_c^s = D_c^{R,s}$~~ | ~~Robotaxi demand = throughput + unserved volume (unit: trips)~~ |
| ~~$U_c^s = H_c^s + L_c^s$~~ | ~~Decompose the outcomes of unserved demand into two types (unit: trips)~~ |

## 4. Notation used in the model

| Symbol | Meaning | Type and unit |
|---|---|---|
| $C$ | Set of potential candidate cities | Input set; cities |
| $S$ | Set of demand scenarios | Input set; scenarios |
| $y_c$ | 1 if city $c$ is entered, else 0 | Decision variable; binary (0 or 1) |
| $q_c$ | Number of vehicles deployed in city $c$ | Decision variable; vehicles (integer) |
| $\kappa_c$ | Number of trips one vehicle can serve during the period | Input parameter; trips/vehicle over the evaluation period. **Operational data or assumptions or fixed?** |
| $Q$ | Total number of vehicles available for deployment | Input parameter; vehicles |
| $\bar{q}_c$ | Upper bound on vehicle deployment in each city | Input parameter; vehicles |
| $F_c$ | Fixed cost of city entry and basic infrastructure (도시c에 진입하는 비용) | Input parameter; currency over the evaluation period |
| $K_c$ | Per-vehicle preparation, deployment, and holding cost over the evaluation period(감가상각비, 보험료, 고정주차 및 관리 비용) | Input parameter; currency/vehicle over the evaluation period |
| $c_c^+$ | Additional loss per unit of unused service capacity (남는 용량 때문에 추가로 발생하는 손실 / 유휴 서비스 용량에 대한 손실 평가계수) | Input parameter; currency/trip of unused capacity |
| $c_c^-$ | Foregone contribution margin per unit of unserved potential demand | Input parameter; currency/unserved trip |
| $M_c$ | Number of **potential ride-hailing usage opportunities** in city $c$ | Input parameter; usage opportunities over the evaluation period. data or assumption |
| ~~$a_c$~~ | ~~1 if operation in the city is feasible at the planning stage, else 0~~ | ~~Assessed using permit, ODD, and facility data. Separate assumptions for unverified conditions~~ |
| ~~$B$~~ | ~~Upper bound on the number of cities that can be entered simultaneously~~ | ~~Company input or an experimental value~~ **~~only when an actual restriction exists~~** |
| ~~$M_c$~~ | ~~Number of **potential ride-hailing usage opportunities** in city $c$, scenario $s$~~ | ~~(Input) Need to be Estimated~~ |
| $p_c^s$ | Probability of choosing our company's robotaxi for that usage opportunity | Scenario input; dimensionless probability in $[0, 1]$ |
| $D_c^s$ | Robotaxi choice demand | Derived input; trips over the evaluation period. $D_c^s = M_c p_c^s$ |
| $D_c^{N,s}$ | Demand not choosing robotaxis (human-driven taxis, other modes, giving up the trip, etc.) | Derived input; trips over the evaluation period. $D_c^{N,s} = M_c(1 - p_c^s)$ |
| ~~$\tau_c^s$~~ | ~~Proportion of unserved customers who switch to human-driven taxis~~ | ~~**estimated by data or fixed?**~~ |
| ~~$R_c^s$~~ | ~~Actual robotaxi throughput~~ | ~~Calculated by the model based on the deployment outcome~~ |
| $O_c^s$ | Unused service capacity | Outcome variable; trips over the evaluation period. $O_c^s = (\kappa_c q_c - D_c^s)^+$ |
| $U_c^s$ | Unserved robotaxi demand | Outcome variable; trips over the evaluation period. $U_c^s = (D_c^s - \kappa_c q_c)^+$ |
| ~~$H_c^s$~~ | ~~Demand that switches to human-driven taxis and is served after being unserved~~ | ~~Calculated using the switching condition below ($H_c^s = \tau_c^s U_c^s$)~~ |
| ~~$L_c^s$~~ | ~~Remaining unserved demand~~ | ~~Groups other modes, giving up the trip, etc. ($L_c^s = (1 - \tau_c^s) U_c^s$)~~ |

※ Demand-related variables consist of the inputs $p_c^s$, $M_c$ and outcome variables calculated from scenario demand and deployment in the model. ※ Outcome variables are not quantities to be predicted by collecting data for each one separately. They are outcomes calculated from demand and deployment.

## 5. Explanations

### A. Construct two types of demand [*Moved from page 3 to page 1 in this revision.*]

$$D_c^s = M_c p_c^s, \qquad D_c^{N,s} = M_c(1 - p_c^s)$$

| Symbol | Meaning | Explanation |
|---|---|---|
| $M_c$ | Number of **potential ride-hailing usage opportunities** in city $c$, ~~scenario $s$~~ | Robotaxi demand + human-driven taxi demand + other demand |
| $p_c^s$ | Probability of choosing robotaxi for that usage opportunity | Need to be specified with reference service conditions such as fares and waiting times |
| $D_c^s$ | Robotaxi choice demand | Demand **before** applying vehicle shortages |
| $D_c^{N,s}$ | Demand not choosing robotaxis | Human-driven taxis, other modes, giving up the trip, etc. |

~~4. Common constraints~~

### B. Constraints

1) Once demand $D_c^s$ and service capacity $\kappa_c q_c$ are determined, the following two values are calculated based on the deployment outcome; they are not variables that need to be investigated separately. $(x)^+ = \max\{x, 0\}$. Both values are measured in trips.

$$\begin{aligned} &~~R_c^s = \min\{D_c^{R,s}, \kappa_c q_c\}~~ && \underline{Actual throughput}, \\ &O_c^s = (\kappa_c q_c - D_c^s)^+ && \text{Remaining service capacity}, \\ &U_c^s = (D_c^s - \kappa_c q_c)^+ && \text{Unserved potential demand}. \end{aligned}$$

~~2) For human fallback, I set the switching proportion $\tau_c^s$ as an input.~~

$$~~H_c^s = \tau_c^s U_c^s, \qquad L_c^s = (1 - \tau_c^s) U_c^s~~$$

~~$\tau_c^s \in [0, 1]$ is the proportion of unserved demand that actually switches to human-driven taxis and is served.~~

~~Then, the cost terms gives~~

$$~~e_{c,H}^- H_c^s + e_{c,L}^- L_c^s = \underbrace{\left[\tau_c^s e_{c,H}^- + (1 - \tau_c^s) e_{c,L}^-\right]}_{\text{Effective shortage cost}} U_c^s.~~$$

~~This is in the case when $\tau_c^s$ is fixed (a constant irrelevant of scenario, city selection and so on).~~

**[B. Constraints has been moved to page 1.]**

~~**※ How to set inputs?**~~

~~I wrote about my thoughts of how I might set the inputs.~~

| Input | How to set it | Use of previously collected data | Data still required |
|---|---|---|---|
| $M_c$ | **Data estimation + market-definition assumptions** | Population, employment, existing ride-hailing, airports, commercial activity | Denominator of the potential travel market. Completed trips alone are insufficient |
| $p_c^{R,s}$ | **Estimation from choice data;** assumptions if unavailable | Candidate explanatory variables such as income, car-free households, and transport accessibility | Robotaxi choice/non-choice, offered fares, waiting times, and alternatives |
| $\kappa_c^s$ | **Estimation from operational data;** assumptions if unavailable | Congestion, roads, weather, charging conditions | Driving, pickup, repositioning, charging, and maintenance times |
| $\tau_c^s$ | **Estimation from switching data;** assumptions if unavailable | Cannot be estimated directly from the current data | Actual behavior after robotaxi demand goes unserved |
| $a_c$ | **Assessment from data + assumptions about launch conditions** | Permit, ODD, and base conditions | Conditions for the company at the planning stage |
| $\bar{q}_c$ | **Facility/operational data or company input** | Charging, parking, and maintenance conditions | Actual scale that can be secured. $Q$ if there is no separate upper bound |
| $Q$ | **Company input or experimental value** | Not estimated from demand data | Number of available vehicles |
| $F_c$ | **Additional cost data or assumptions** | Some base and land costs | Initial construction and entry preparation costs |
| $K_c$ | **Additional cost data or assumptions** | Cannot be determined.. | Vehicle deployment and holding costs for the evaluation period |
| $c_c^+$ | **Company loss data or assumptions** | Cannot be determined.. | Burden of remaining capacity. No double counting with $K_c$ |
| $c_{c,H}^-, c_{c,L}^-$ | **Company loss data or assumptions** | Cannot be determined.. | Differences in opportunity loss and customer churn by switching type |

※ $y_c, q_c$ are not included; values determined by the model

## 5. Driving factors

### A. Driving factors of $M$

The papers below mainly explain **observed ride-hailing usage**, so they do not include the 'unrealized demand' included in the definition of $M$, making direct estimation impossible. However, I suppose the demand patterns provided by these papers lead to constructing 'potential travel opportunities $M$' through modeling.

| Variable | Supporting paper | Scope established in the paper | Measurement data |
|---|---|---|---|
| **Population and population density** | Elkhouly and Alhadidi, *Ride-sharing Determinants* (paper); Zhang et al., *Examining spatial heterogeneity⋯* (paper) | Relationship of population and local density with conventional ride-hailing usage. Density effects may vary by area | ACS |
| **Employment and economic activity** | Elkhouly and Alhadidi (paper) | Uses the economically active population and employment environment to explain usage | LEHD/LODES, ACS |
| **Income** | Elkhouly and Alhadidi (paper) | Analyzes the relationship between income and usage in Chicago | ACS B19013 |
| **Share of car-free households** | Elkhouly and Alhadidi (paper) | Analyzes the relationship between households without cars and usage | ACS B08201 |
| **Land-use mix, accessibility, and commercial activity** | Zhang et al., *Examining spatial heterogeneity⋯* (paper) | Nonlinear and spatial differences in the urban environment and ridesourcing usage | EPA SLD |
| **Restaurant and leisure-facility density** | Loa, Lee, and Circella, *Just another ride-hailing?* (paper) | Relationship with **robotaxi trip counts in SF**. Does not separate the effects of $M$ and $p$ | Census CBP, OpenStreetMap |
| **Public transport and walking environment** | Zhang et al. (paper); Cats et al., *Beyond the dichotomy⋯* (paper) | Analyzes the local environment and travel-time competitiveness and substitutability with public transport. Not a uniform increase/decrease effect | EPA SLD, GTFS |
| **Airport-related market size** | Zhang et al. (paper) | Analyzes airport travel as a separate spatial category. **Does not estimate the direct effect of airport passenger counts** | FAA passenger data. Used as a proxy for market size |
| **Time, day of week, and season** | Elkhouly and Alhadidi (paper) | Analyzes differences in usage over time. Specific variables such as holidays are selected through separate design | Dates and times in trip data |
| **Weather** | Additional: Liu, Jiang, and Chen, *Quantifying the impact of weather on ride-hailing ridership: Evidence from Haikou, China* (2021, paper) | Analyzes the relationship between weather and ride-hailing usage. Effects in U.S. cities require revalidation | NOAA CDO |

## B. Driving factors of $p$

| Variable | Supporting paper | Scope of application | Required measurement data |
|---|---|---|---|
| **Relative fare** | Additional: Yin and Cherchi, *Willingness to pay for automated taxis*⋯ (2024; online 2022, paper) | Choice experiment between conventional taxis and automated taxis | Offered fares for both + choice outcomes |
| **Relative waiting time** | Yin and Cherchi (paper) | Analyzes choices under different service conditions | Offered waiting times for both + choice outcomes |
| **In-vehicle travel time** | Yin and Cherchi (paper) | Analyzes choice trade-offs between fares and time | Offered travel times for both |
| **Trust, price value, and usage habits** | Additional: Wei et al., *Using the Extended Unified Theory*⋯ (2024, paper) | Study of robotaxi acceptance intention and behavior. **Survey scales cannot be interpreted directly as choice probabilities** | Individual attitude and habit surveys |
| **Age and robotaxi-related knowledge** | Yin and Cherchi (paper) | Choice differences by individual characteristics. Fixing the sign, such as "younger people always prefer it, is inappropriate | Link individual characteristics to choice data |
| **User reviews and in-vehicle support features** | Yin and Cherchi (paper) | Analyzes customer ratings and features such as destination changes and communication with an operator as choice attributes | Service attributes in the choice experiment |
| **Experience using robotaxis** | Additional: Yin, Cherchi, and Nettle, *Living in a Ci ty Where Automated Taxis are Operating and Using Them*⋯ (2023, paper) | Analyzes preference differences by whether service operates in the residential area and by individual usage experience | Individual usage experience |
| **Competitiveness of alternatives such as public transport** | Cats et al., *Beyond the dichotomy*⋯ (paper) | Evidence from **conventional ride-hailing**. Not a robotaxi-specific preference effect | Availability, fares, and times of alternatives |

~~**6. Things needed to be clarified**~~

~~1. how to set scenarios?($M$,$p$,$\tau$, $\kappa$)~~

~~-> Kannan's paper?~~

~~2. What can be estimated from the existing research materials?~~

~~3. What additional data are needed?~~

~~4. How should assumptions be made for symbols for which data are unavailable?~~

~~5. Can a solution be obtained for the objective-function formulation?~~

~~6. If data cannot be obtained for the remaining symbols other than the outcome variables that need to be estimated, how should they be set?~~

~~7. How to choose which factors to use for model?~~